

CSL-402

Professional Practices

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Topic # 1: (Chapter # 1)

The Engineering Profession

Profession: (A paid occupation, especially one that involves prolonged training and a formal qualification)

⇒ A profession is a job, what you do for a living.

⇒ An occupation or career is also called profession.

⇒ A profession is a disciplined group of individuals who adhere to ethical standards and possess specialized knowledge and skills in a specific field.

⇒ In other words:

A profession is a special kind of job that needs a lot of study and training.

⇒ Professionals: People who do this job are called professionals. They know a lot about their work and are really good at it.

Key characteristics of a profession:

Specialized knowledge:

Requires advanced education and training.

⇒ The profession is organized into one or more professional bodies.

⇒ The profession lays down the standards of conduct with which its members must comply.
Ethical standards:

Adherence to a code of conduct.

Public service orientation:

Focus on serving the needs of others.

Autonomy:

Professionals have independence in decision making

Licensing or certification:

Often required to practice

Example of professions:

- Doctors
- Lawyers
- Engineers
- Accountants
- Architects
- Teachers etc

Professional: (Profs)

⇒ A professional is a member of a profession.

⇒ Someone who is a member of a profession that requires special training, education & skill.

⇒ A professional is:
• Competent

- Efficient
- Experienced
- Licensed
- Qualified
- Skillful

Professional standards/Qualities:

- ⇒ Accountability
- ⇒ Confidentiality
- ⇒ Honesty
- ⇒ Integrity
- ⇒ Law-abiding
- ⇒ Loyalty
- ⇒ Objectivity
- ⇒ Transparency

Professionalism:

⇒ The competence or skill expected of a professional

⇒ Professionalism is behaving in a way that shows you are skilled, responsible and respectful at work.

⇒ It means:

- Doing your job well and on time.
- Being polite and helpful to others.
- Dressing and acting appropriately for workplace.
- Always trying to improve your skills.

Professional Practices:

- ⇒ Professional practices are the specific skills, knowledge, and behaviors used by people in a particular job.
- ⇒ It's how professionals apply their expertise in real-world situations.
- ⇒ Professional practices are things that someone does at work to be a good worker.

So, a profession is what someone is trained to do, and professional practices are how they actually do it.

Professional Bodies:

- ⇒ Professional bodies are organizations that represent and support people working in a specific profession.
- ⇒ They set standards for the profession, provide training and development opportunities, and protect the public interest.

⇒ A professional body usually starts by a group of people coming together because of shared interest in a particular type of activity.

Key roles of professional bodies

- ⇒ Setting standards
- ⇒ Providing education and training to help professionals stay up-to-date.
- ⇒ Representing the profession
- ⇒ Protecting the public

Examples:

- BCD was set up in 1957 ^{→ Learned society of ACM.}
- IEEE was set up in 1946 ^{→ focus on engineering and design.}
- ACM was set up in 1947 ^{→ focus on math and science.}

Note: A professional body may also be started by people engaged in the same type of activity, who want to protect their business against others who may be trying to enter it without having the proper knowledge or who may be practising it dishonestly.

(Note: Structure of Engineering profession on Page #)

Q. Explain Structure of engineering Profession?

⇒ Engineering profession means a field that have linked with scientific field.

[Rough intro here!]
⇒ Engineering institutions:

Chartered professional bodies, cover single or closely related disciplines; institute of electrical engineers, Institute of mechanical engineers, BCS

work in UK and govern all related degree to computers, set rules & regulations.

○ In Pakistan, have Pakistan engineering Council for engineering programs, Higher education Commission regulates computer degrees.

○ These bodies protect interests of engineers. Are consulted by government as source of expert and impartial advice. e.g BCS & HEC can be consulted for data protection, legal protection of software, misuse of computers.

They also have role in advancement knowledge in their field.

- ⇒ An engineer is qualified to:
- Develop new tech
 - Apply existing tech in new ways
 - Take responsibility for large high risk projects

Two-tier Structure:

⇒ The engineering profession has two tier structure

First level : Engineering institutions:

⇒ These are professional organizations that oversee specific engineering fields. They set standards, promote professional development & represent the interests of engineers.

- IEEE
- IME
- University of Engineering & Technology.

Second Level: Engineering Council

⇒ This is a higher level body that recognizes & represents engineering institutions, as its "Nominated bodies".

⇒ It ensures that these institutions maintain high standards & ethical practices.

Ex^o

- Pakistan engineering Council

⇒ Engineering Council acts as "umbrella" body and represents the interest of engineering profession.

Shoot Questions (Related to previous Topic)

Q. List down five things that define a true professional?

- o Put customer satisfaction first
- o make expertise your specialty
- o Do more than expected
- o Do what you say and say what you can do
- o Communicate effectively
- o Praise your peers not yourself
- o Share your knowledge
- o Follow exceptional guiding principles
- o Share knowledge
- o Say Thank you
- o Keep a smile on face

Q. what is the engineering profession?

⇒ is a broad field that applies scientific and mathematical principles to design, build and maintain structures, machines, systems and processes. Engineers solve problems and find efficient solutions to meet human needs. Focuses on physical systems, structures, and processes.

Q. What is computing profession?

- ⇒ Encompasses individuals who work with computer systems and networks to process, store and transmit information.
- ⇒ it involves a wide range of activities from developing software and hardware to managing databases and ensuring network security.
- ⇒ Primarily deals with software, data and information systems.

Q. What is code of ethics?

Set of rules for professionals in conducting business with honesty & integrity.